

Cup Course Overview

A calculus and vector treatment of basic physics. This is the first semester, which includes mechanics of motion, vibration and sound. Required of physics majors. Offered every fall.

Prerequisite or Concurrent Course: Math 211 · **Credits:** 4

Target Student Learning Outcomes

- Analyze vectors and apply them in multi-dimensional kinematics and dynamics.
- Use calculus to model and solve mechanics problems.
- Apply Newton's laws and mechanics principles to describe and predict motion.
- Solve problems involving energy, momentum, rotational motion, oscillations, and waves.

Full day-by-day SLOs: see the companion course website.

Calendar Weekly Rhythm

Mon/Wed/Fri — New content, with a pre-class Reading check due before each meeting

Tuesdays — Lab session · **First day of week** — Weekly Check-In Quiz

Sandglass When Assignments Are Due

- Daily Assignments:** before the start of the next class
- Complex Problems** (chapter-wise): Moodle by the following week's Sunday, 11:59 p.m.; write-up in class the next meeting
- Concept Quest:** biweekly, last day of the week

Calendar Key Dates

Classes begin: Mon, Aug 31 · **Labor Day:** Mon, Sep 7 (no class)

Community Engagement Day: Mon, Sep 21 (no class)

Homecoming: Oct 2–4 · **Mid-term break:** Oct 19–21

Last day to withdraw: Fri, Oct 30

Thanksgiving break: Nov 25–27

Last day of classes: Fri, Dec 4 · **Reading Day:** Mon, Dec 7

Final exams: Dec 8–10

Per Goshen College Traditional Academic Calendar, 2026–27.

Bar Chart Grade Distribution



Letter grade scale: A 93+ · A– 90+ · B+ 87+ · B 83+ · B– 80+ · C+ 77+ · C 73+ · C– 70+ · D 60+ · F <60. All grades posted in the Moodle gradebook.

Pencil Assignment Types & Deadlines

Type	Deadline
Pre-class Reading	Before class — 2% of grade
Daily Assignments	Before the start of the next class — 6% of grade
Complex Problems	Chapter-wise, not per day: Moodle by the following week's Sunday at 11:59 p.m., write-up due in class the next meeting — 6% of grade
Weekly Check-In Quiz	Beginning of class, first day of the week — 5% of grade
Concept Quest	Biweekly, last day of the week — 10% of grade
Labs	Every Tuesday — 15% of grade
Presentation	Date announced in Moodle — 5% of grade
Community Activities	Dates announced in Moodle — 1% of grade

Calendar Tentative Exam & Test Schedule

Exam	Date & Coverage
Test 1	Tuesday, October 6 (lab period) — covers units and dimensions, 1D motion, vectors, and 2D motion — 5% of grade
Test 2	Tuesday, November 3 (lab period) — covers Newton's laws, momentum and collisions, and work and energy — 10% of grade
Test 3	Tuesday, November 17 (lab period) — covers material through rotational motion — 15% of grade
Final Exam	December 8–10 (exact date/time set by the registrar's final exam schedule) — comprehensive — 20% of grade

Test dates are tentative and may shift slightly with notice; confirmed dates are posted in Moodle and the course schedule.

Calendar Weekly Topic Schedule

Week	Dates	Topic	Week	Dates	Topic
1	Aug 31	Units & Dimensions	9	Oct 26–30	Work & Energy
2–3	Sep 7*–18	1D Kinematics & Free Fall (*no class Sep 7)	10–11	Nov 2–13	Rotational Motion · Test 2, Tue Nov 3
4	Sep 21*–25	Vectors (*Community Engagement Day)	12	Nov 16–20	Oscillations (SHM) · Test 3, Tue Nov 17
5	Sep 28–Oct 2	2D Motion & Projectiles	13	Nov 23–27*	Waves (*break Nov 25–27)
6–7	Oct 5–16	Newton's Laws & Friction · Test 1, Tue Oct 6	14	Nov 30–Dec 4	Sound & Doppler Effect; Review
8	Oct 19*–23	Momentum & Collisions (*break Oct 19–21)	—	Dec 8–10	Final Exam period